26900397

Discipline Specific Core Course (DSC) MG5DSCPES300 – PHYSIOLOGY OF EXERCISE

MGU-UGP (Honours) | BPES | Fifth Semester | October 2026

ANSWER KEY

PART A – MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

1. What major physiological discovery was made by William Harvey in 1628?

b) The continuous circulation of blood.

2. Because the body stores very little raw ATP, what must cells constantly do?

c) Recycle millions of ADP molecules back into ATP.

3. Which of the following is NOT an adaptation of skeletal muscle to regular exercise?

c) Decreased capillary density.

4. Why are there numerous mitochondria in the presynaptic terminal?

b) To provide the energy needed for neurotransmitter release.

5. Skinfold calipers measure:

b) Subcutaneous fat thickness.

6. Body fat percentage >25% in men is classified as:

c) Overfat.

7. How does trained respiratory efficiency affect breathing at rest?

b) Breathing rate is reduced because each breath is deeper and more efficient.

8. What causes the initial feeling of fatigue before a 'second wind' occurs?

b) An initial oxygen deficit as the cardiovascular system takes time to adjust.

9. Which exercise approach is best for long-term BP control?

c) Regular moderate exercise.

10. Which metabolic condition is commonly associated with PCOD?

b) Insulin resistance.

PART B – SHORT ANSWER QUESTIONS (ANSWER ANY 5; 2 MARKS EACH)

11. Describe the molecular structure of ATP.

- ATP stands for adenosine triphosphate, the immediate energy-carrying molecule used by cells.
- It is made up of adenine (a nitrogenous base), ribose (a five-carbon sugar), and three phosphate groups.
- When the terminal phosphate bond is hydrolysed, ATP is converted into ADP and inorganic phosphate, releasing usable free energy.
- This energy supports muscle contraction, active transport across cell membranes, nerve function, and many cellular reactions.

12. What happens to the pyruvic acid produced during glycolysis if no oxygen is present?

- When oxygen availability is insufficient relative to energy demand, pyruvate in human muscle cells can be converted into lactate.
- This reaction regenerates NAD^+ from NADH, allowing glycolysis to continue.
- Glycolysis can therefore continue to produce a small amount of ATP without directly requiring oxygen, although it yields much less ATP per glucose molecule than aerobic respiration.

13. Give an example of heat loss via radiation.

- Radiation is the transfer of heat from the body to cooler objects in the surroundings through electromagnetic waves, without direct contact.
- For example, a person may lose heat from the skin to cooler walls or objects in a room. Radiation is one of the body's mechanisms of heat exchange, alongside convection, conduction, and evaporation.

14. Name two hormones that significantly influence baseline metabolism.

- Thyroid hormones, particularly thyroxine (T4) and triiodothyronine (T3), are major regulators of basal metabolic rate.
- These hormones influence cellular metabolic activity and oxygen consumption. Other hormones, including catecholamines, also influence energy expenditure; insulin is particularly important in the regulation and use of nutrients.

15. Define fat mass and explain its role in the body.

- Fat mass is the total amount of fat tissue in the body, including essential fat and stored fat.

- It acts as an energy reserve, provides thermal insulation, cushions and protects internal organs, and contributes to hormone production and normal physiological functions.
- Excessive fat accumulation, particularly around abdominal organs, may increase the risk of metabolic and cardiovascular diseases.

16. Explain why exercise prescription should be individualized for people with chronic diseases.

- People with chronic diseases differ in diagnosis, severity, symptoms, medication, age, fitness, and functional ability.
- Individualized exercise helps select safe types, intensity, duration, and frequency of activity, while reducing the risk of injury or worsening symptoms.
- Progress should be gradual and monitored; medical assessment or professional supervision may be necessary depending on the condition.

17. Justify the need to consider individual variation when studying sex differences.

- People vary in body size, body composition, hormones, genetics, age, training history, nutrition, and fitness, even within the same sex.
- If these factors are ignored, a difference attributed to sex may actually reflect other variables or overlap between individuals.
- Considering individual variation supports fairer comparisons, more accurate conclusions, and exercise programmes based on personal needs rather than stereotypes.

PART C – SHORT ESSAY QUESTIONS (ANSWER ANY 4; 5 MARKS EACH)

18. Outline five current sub-disciplines of modern physiology and explain their primary focus.

Physiology is the scientific study of how living organisms and their parts function. Modern physiology has several specialized sub-disciplines.

1. Cellular physiology: examines cell membranes, transport, metabolism, signalling, and how cells maintain their internal environment.

2. Neurophysiology: studies the functioning of the brain, spinal cord, nerves, and synapses, including nerve impulses and sensory and motor control.

3. Cardiovascular physiology: studies the heart's pumping action, blood flow, blood pressure, and the transport of oxygen and nutrients.

4. Respiratory physiology: studies ventilation, gas exchange in the lungs, transport of oxygen and carbon dioxide, and regulation of breathing.

5. Exercise physiology: examines acute body responses to exercise and long-term adaptations to training, including changes in cardiovascular, respiratory, muscular, and metabolic function.

Other recognized fields include renal, endocrine, gastrointestinal, reproductive, and environmental physiology.

19. Summarize the holistic importance of metabolism in the human body, detailing how it supports growth, tissue repair, body temperature regulation, and overall health.

- Metabolism is the sum of chemical reactions in the body. It includes catabolism (breakdown of molecules to release energy) and anabolism (building molecules using energy).
- It breaks down carbohydrates and fats to produce ATP, which powers muscle contraction, nerve activity, membrane transport, and cellular synthesis.
- Anabolic processes use amino acids, glucose, fatty acids, minerals, and other nutrients for growth and development of new cells and tissues.
- Metabolism provides energy and raw materials for protein synthesis, tissue repair, and restoration of damaged structures after exercise or injury.
- Metabolic reactions release heat, helping maintain body temperature along with circulation, sweating, and behavioural responses.
- It supports organ function, immunity, brain activity, physical performance, and maintenance of internal balance.
- Nutrition, hormones, age, physical activity, sleep, and disease influence metabolic rate and pathways.

20. Describe the structural components of the neuromuscular junction (NMJ) and explain how the destruction of acetylcholine receptors on the motor end plate would affect muscle contraction.

- The neuromuscular junction (NMJ) is a specialized synapse between a motor neuron and a skeletal muscle fibre.
- Presynaptic terminal: Contains synaptic vesicles filled with acetylcholine (ACh). Nerve impulses trigger calcium entry and ACh release.
- Synaptic cleft: The narrow gap across which ACh diffuses; acetylcholinesterase breaks down ACh.
- Motor end plate: A specialized muscle membrane region containing numerous nicotinic ACh receptors.

- Muscle contraction: ACh binds to receptors, generating an end-plate potential and muscle action potential, which triggers calcium release and actin–myosin interaction.
- Effect of receptor destruction: Reduced ACh receptor availability prevents sufficient stimulation of the muscle membrane, reducing or preventing muscle action potentials and calcium release.
- Result: Muscle contraction becomes weak or fails completely, causing muscle weakness, fatigue, or severe paralysis.

21. Compare lean body mass adaptations to resistance training versus endurance training.

- Lean Body Mass: Body mass excluding fat, including muscles, bones, organs, and body water.
- Resistance Training: Increases muscle protein synthesis, muscle size, strength, power, and connective-tissue capacity.
- Endurance Training: Improves mitochondrial content, oxidative enzyme activity, capillary supply, and fatigue resistance, with generally smaller increases in muscle size.
- Functional Benefits: Resistance training improves maximal force and muscle hypertrophy, while endurance training improves aerobic capacity and sustained performance.
- Common Benefits: Both improve body composition, health, and physical function.
- Influencing Factors: Adaptations depend on training intensity, duration, programme design, nutrition, recovery, age, genetics, and training status.
- Combined Training: Combining resistance and endurance training develops a broader range of fitness benefits.

22. Explain how the principles of progression and overload can be applied to develop strength and muscular fitness in male and female students.

- Overload: Apply a training load greater than the body is accustomed to, to improve strength and muscular fitness.
- Initial Assessment: Assess students' health, experience, technique, and current strength before starting training.

- Progression: Gradually increase resistance, repetitions, sets, range of motion, or exercise difficulty as students adapt.
- Balanced Programme: Train major muscle groups with proper technique, suitable rest intervals, and adequate recovery.
- Monitoring: Record loads, repetitions, fatigue, and progress; modify training if pain or poor technique occurs.
- Individualization: Apply the same principles to male and female students according to their individual abilities, goals, experience, and responses to training.

23. Explain aerobic exercise prescription for inactive older adults.

Aerobic exercise prescription for inactive older adults aims to improve cardiovascular endurance, mobility, daily functional ability, and overall health while ensuring safety.

- Frequency: Begin with about 3 days per week and gradually increase activity to most days.
- Intensity: Start at light intensity and progress to moderate intensity as tolerated, using the talk test to monitor effort.
- Time: Start with 5–10-minute sessions and gradually increase duration. Short sessions can be accumulated throughout the day.
- Type: Choose low-impact activities such as walking, stationary cycling, or water-based exercise.
- Safety: Include warm-up and cool-down, proper footwear, hydration, and a safe environment. Stop if chest pain, faintness, or unusual breathlessness occurs.
- Progression: Increase duration and intensity gradually according to fitness, health, and tolerance.
- Long-term Goal: Aim for at least 150 minutes of moderate-intensity aerobic activity per week, when medically appropriate. Seek medical advice or supervised exercise when necessary.

PART D – ESSAY QUESTIONS (ANSWER ANY 2; 15 MARKS EACH)

24. Catabolism vs. Anabolism: Define human metabolism and clearly distinguish between its two primary phases, providing examples for each.

- Human metabolism is the sum of all chemical reactions in the body that maintain life, produce energy, support growth, and repair tissues.
- Catabolism: Breaks down complex molecules into simpler substances and releases energy in the form of ATP.
- Examples of Catabolism: Glucose breakdown, glycogenolysis, fat oxidation, and protein breakdown.
- Importance of Catabolism: Provides energy for muscle contraction, nerve activity, and organ function.
- Anabolism: Builds complex molecules from simpler substances and requires energy, usually supplied by ATP.
- Examples of Anabolism: Protein synthesis, glycogenesis, lipid synthesis, and tissue growth and repair.
- Importance of Anabolism: Supports growth, recovery, tissue repair, and energy storage.
- Differences: Catabolism breaks down molecules and releases energy, whereas anabolism builds molecules and requires energy.
- Regulation: Both processes are influenced by hormones, nutrition, physical activity, age, and the body's needs. Insulin promotes nutrient storage, while glucagon helps mobilize stored energy.
- Relationship: Catabolism supplies energy during exercise, while anabolism supports recovery and replenishment of energy stores.
- Conclusion: Both processes work together to maintain energy balance, growth, tissue repair, and normal body functions.

25. Acute Cardiovascular Responses: Analyze the short-term cardiovascular responses to a single bout of exercise, specifically explaining changes in heart rate, stroke volume, and cardiac output.

- A single bout of exercise increases the demand for oxygen and nutrients in working muscles, requiring immediate cardiovascular adjustments.

- Heart Rate (HR): Increases due to reduced parasympathetic activity and increased sympathetic stimulation. It rises with exercise intensity and gradually decreases during recovery.

- Stroke Volume (SV): The amount of blood pumped per heartbeat increases due to greater venous return, improved ventricular filling, and increased cardiac contractility. It may plateau at higher intensities.
- Cardiac Output (Q): The volume of blood pumped per minute increases as heart rate and stroke volume rise. It is calculated as:

Cardiac Output = Heart Rate × Stroke Volume

- Blood Pressure: Systolic blood pressure generally increases during dynamic exercise, while diastolic pressure changes little or may decrease slightly.
- Blood Flow Distribution: Blood flow to active muscles increases, while flow to some less-active organs may decrease during strenuous exercise.
- Venous Return: The skeletal muscle pump and respiratory movements increase blood return to the heart, supporting stroke volume.
- Temperature Regulation: Skin blood flow may increase to help dissipate heat.
- Recovery: After exercise, heart rate and cardiac output gradually return toward resting levels.
- Conclusion: These acute responses improve oxygen delivery, nutrient supply, waste removal, and temperature regulation. Their magnitude depends on exercise intensity, fitness, age, hydration, and health status.

26. Evaluate the relationship between BMI and morbidity and mortality risk. Discuss the evidence for BMI as a predictor of health outcomes.

- Body Mass Index (BMI) is calculated as weight (kg) ÷ height² (m^2). It is a screening tool for assessing adult weight status, not a direct measure of body fat.
- BMI and Morbidity: High BMI, especially obesity, is associated with type 2 diabetes, hypertension, cardiovascular disease, certain cancers, osteoarthritis, and sleep apnoea.
- Low BMI: May indicate undernutrition, reduced muscle mass, nutritional deficiencies, and increased vulnerability to illness.
- BMI and Mortality: Studies show increased mortality risk at very low and very high BMI levels. However, age, smoking, existing diseases, fitness, and unintentional weight loss can influence these associations.

- Evidence Supporting BMI: It is simple, inexpensive, reproducible, and useful for population studies, routine screening, and monitoring weight changes over time.
- Limitations: BMI cannot distinguish fat from muscle, measure abdominal fat distribution, or fully reflect an individual's fitness and metabolic health.
- Complementary Assessment: BMI should be combined with waist circumference, blood pressure, blood glucose, lipid profile, medical history, physical activity, and body-composition assessment when appropriate.
- Conclusion: BMI is a useful screening tool associated with morbidity and mortality risk, but it should not be used alone to diagnose health conditions or predict individual health outcomes.

27. Critically assess the influence of food habits and physical inactivity on chronic bronchitis.

- Chronic bronchitis is a long-term inflammation of the bronchi characterized by productive cough for at least three months in two consecutive years. Smoking is a major cause, while diet and physical inactivity influence overall health and symptoms.
- Balanced Nutrition: A diet rich in fruits, vegetables, whole grains, protein, vitamins, and minerals supports immunity and tissue repair.
- Poor Dietary Habits: Inadequate nutrition can weaken immunity, reduce recovery, and contribute to muscle weakness.
- Undernutrition and Obesity: Muscle loss can weaken respiratory muscles, while excess body weight may increase the work of breathing.
- Hydration: Adequate fluid intake supports general health and may help maintain comfortable airway secretions.
- Physical Inactivity: Reduces cardiovascular fitness, muscle strength, and exercise tolerance, increasing fatigue and breathlessness during daily activities.
- Cycle of Inactivity: Breathlessness discourages exercise, leading to deconditioning and further difficulty with physical activity.
- Benefits of Exercise: Walking, cycling, strength training, and pulmonary rehabilitation can improve endurance, muscle function, independence, and quality of life.
- Other Risk Factors: Smoking, air pollution, and occupational irritants are important causes. Avoiding these exposures and receiving appropriate medical care are essential.

- Conclusion: Balanced nutrition, adequate hydration, regular suitable exercise, avoidance of irritants, and medical treatment help manage symptoms and improve quality of life, but do not guarantee a cure.